

Resolution Geometry

Uncertainty Geometry: Indistinguishability Is the Primitive; Probability, Gaussian Propagation and Monte Carlo Are Theorems

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Abstract

Earlier attempts built uncertainty from Monte Carlo, which assumes the probabilistic structure it should explain. This paper starts from the primitive of Resolution Geometry (RG): the indistinguishability of a declared experiment. That is the kernel of the observation law at the exact layer, and Hellinger distinguishability with its infinitesimal form, the Fisher metric, at finite resolution.

Probability enters only through three locks, each a theorem with a stated hypothesis:

- exchangeability of records (de Finetti);
- mixing dynamics, which forget their input ensemble and converge to an invariant law;
- quantum structure, which forces the trace (Born) rule and forbids noncontextual hidden values.

Statistics is locked separately. The Fisher metric is forced (Čencov), and Gaussian error propagation is the locally asymptotically normal (LAN) limit on regular strata. Priors are forgotten on the quotient (Bernstein–von Mises) and *never* along the observational kernel.

Non-circularity becomes a computable certificate. An uncertainty statement is FORCED iff it is insensitive, relative to its own uncertainty scale, over a declared admissible family of inputs, and CONDITIONAL otherwise. Monte Carlo is objective exactly on FORCED outputs, and valid only with an effective-sample-size certificate.

The early CHI coherence-volume formula is audited. Its bottleneck inequality is reversed, its Fisher-rank seal is blind to misspecification, and density cancels from it although power is proportional to density. Its condition-number hazard is retained as a rank-wall indicator. A ten-check battery passes 10/10 on five seeds, and three canaries are caught.

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1 The question

What is uncertainty before any probability is assumed? And when, and why, does it become a probability distribution that Monte Carlo can sample? An answer must satisfy a hard requirement: *no input may be smuggled*. Any assumption about input distributions, error laws or priors must either provably drop out of the conclusion or be reported as a declaration. The paper derives both directions and tests them (UG01–UG10).

2 Audit of the early attempts

Monte Carlo first. Sampling from an assumed input law and propagating it through a model returns a distribution. It does not show which features of that distribution come from the data or the dynamics and which come from the assumption. Section 7 turns this into a test.

The CHI coherence volume (UG09). The formula $\chi = Mv^2/(\Phi gh\rho)$ was an early geometric compression of drive, constraint and resistance. Under test:

- *Bottleneck theorem, reversed.* The harmonic mean satisfies $\min \chi \leq \chi_{\text{eff}} \leq \bar{\chi}$. The claim $\chi_{\text{eff}} \leq \chi_{\text{min}}$ is false; 1000/1000 random cases lie between the minimum and the mean.
- *The Fisher-rank seal is blind to misspecification.* The Fisher information of $Y = k\chi$ in k has rank 1 at every operating point. Meanwhile a v^2 law calibrated at 20 m/s misses rolling-plus-drag fuel power by 18% at 10 m/s and 22% at 40 m/s. Constant rank certifies identifiability of k , not correctness of the law.
- *Density cancels.* With $M = \rho Av$ in the wind example, χ is independent of ρ , while turbine power is proportional to ρ ; the error at altitude ($\rho = 0.9$) is 36%.
- *Retained.* The hazard $H = \lambda_{\text{min}}/\lambda_{\text{max}}$ of the Fisher matrix tends to 0 at a rank wall (1, 10^{-2} , 10^{-4} for decreasing gain). It was the right instinct, and it is the precursor of the rank certificate of the ladder (L38).

3 The primitive

Definition 3.1 (Indistinguishability). For a declared experiment $F : X \rightarrow Y$:

- at the exact layer, two states are indistinguishable iff $F(x) = F(x')$, which is the kernel $\mathcal{K}$ (L1);
- at finite resolution, indistinguishability is the tolerance relation $H(F(x), F(x')) \leq \varepsilon$. It is reflexive and symmetric but not transitive (Poincaré’s paradox), and it is the honest form of “within measurement error”;
- infinitesimally, it is the Fisher metric, forced by Markov invariance (Čencov; UG05).

Theorem 3.2 (Uncertainty splits along the quotient; UG03). *Let q be any requested quantity. Then:*

- its component that factors through the quotient Q is constrained by data;
- its component along the fibres of $\mathcal{K}$ is never constrained by data, so its uncertainty is exactly what was declared.

In the linear-Gaussian case, the posterior along the kernel tends to the prior's conditional variance on the fibre, while identifiable components forget the prior at rate $1/n$.

Proof. The likelihood is constant on fibres (L2), so Bayes' rule multiplies the prior by a function of $\pi(x)$ only. The conditional prior on each fibre is therefore unchanged. On the quotient, Bernstein–von Mises applies under regularity [3]. $\square$

Measured with two priors: the identifiable difference scales as 0.10, 0.0095, 0.00098 and 0.000098 for $n = 10, \dots, 10^4$, while the kernel-variance difference stays at 0.312.

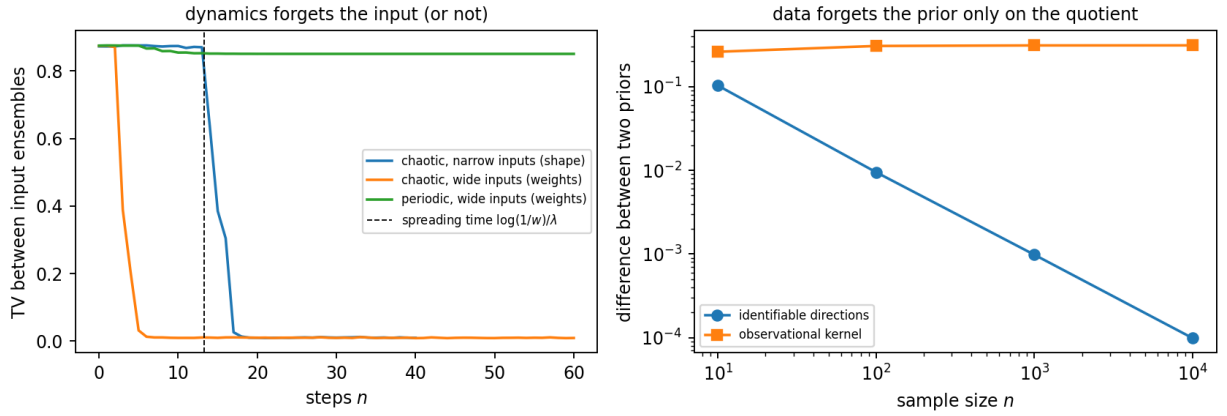


Figure 1: Left: input ensembles are forgotten by chaos (shape and weights) after the spreading time, but not by a periodic system (phase weights persist). Right: data forget the prior on the quotient (slope -1) and never along the observational kernel.

4 The probability lock

Probability is not a primitive of RG. It is forced in exactly three ways.

Imported Theorem 4.1 (de Finetti; UG01; [4]). *An infinite exchangeable record is a mixture of i.i.d. laws; its frequencies converge almost surely.*

A Pólya urn (exchangeable) has a late frequency spread of 0.009. A deterministic record of doubling blocks (non-exchangeable) oscillates forever between $\frac{1}{3}$ and $\frac{2}{3}$ (spread 0.33). Without exchangeability, no probability is forced.

Theorem 4.2 (Mixing forces probability; UG02; [8]). *For a mixing system with $\lambda > 0$, any input ensemble of width w is forgotten after the spreading time $\log(1/w)/\lambda$ plus a mixing time, and the forecast law converges to the invariant (physical) measure. For a contracting periodic system the input's mass split over the cycle phases is preserved forever.*

For the logistic map at $r = 4$, three input laws of width 10^{-4} start at TV distance 0.87, fall below 0.013 after about 17 steps, and reach the arcsine law to TV 0.006. Wide inputs are forgotten in about 6 steps. At $r = 3.5$, wide inputs stay at TV 0.85 forever. This is when Monte Carlo over initial conditions is justified: after the dynamics has forgotten the declaration. It connects the uncertainty geometry to the chaos frontier law of the companion paper.

Imported Theorem 4.3 (Gleason, Busch; UG07; [5, 6]). *In dimension ≥ 3 (and for POVMs in every dimension), noncontextual probabilities on quantum events are $\text{Tr}(\rho E)$. Quantum probability is locked by structure: the trace rule sums to 1 in every basis (error 0), while a squared rule varies from basis to basis by 0.20.*

5 The statistics lock

Imported Theorem 5.1 (Statistics lock; [1, 2, 3]). • *The Fisher metric is the unique Markov-invariant metric (UG05: Fisher change 0, Euclidean change 0.35).*

- *Regular experiments are locally asymptotically normal: the experiment is locally a Gaussian shift with covariance I^{-1}/n .*
- *Under the same regularity, posteriors forget priors.*

Theorem 5.2 (Where Gaussian propagation is correct; UG04). *Linearized (Gaussian) error propagation is the LAN limit. It is correct on regular strata of the quotient and fails at walls. At a fold, where b and $2d-b$ give the same law, the linearized interval selects one branch. The RG uncertainty set, the preimage of the observed interval (a correspondence, L19), keeps nominal coverage.*

On the regular branch the coverages are: linearized 0.947, RG set 0.947. On the other branch of the fold: linearized 0.000, RG set 0.94–0.96 across seeds.

6 The quantum lock

Theorem 6.1 (Quantum uncertainty is not a latent kernel; UG06; [7]). *No assignment of non-contextual values reproduces quantum observables:*

- *0 of the 512 assignments of ± 1 satisfy the Peres–Mermin constraints, which the Pauli-product observables satisfy exactly;*
- *CHSH gives 2 classically and $2\sqrt{2}$ quantum mechanically.*

Quantum uncertainty therefore cannot be represented as ignorance over a latent kernel of hidden values. It is locked by the trace rule, and its metric module is fixed by the declared task (SLD, BKM or WY; Quantum Metric Selection).

7 Non-circularity

Hypothesis 7.1 (H-adm). Before the output is seen, the protocol declares the admissible family $\mathcal{A}$ of input hypotheses: priors, error laws, or initial ensembles.

Definition 7.2 (Non-circularity certificate). An uncertainty statement S is FORCED iff its spread over $\mathcal{A}$ is at most 0.1 times its own uncertainty scale. Otherwise it is CONDITIONAL, and it is reported together with the declaration it depends on.

Theorem 7.3 (No smuggling). *Under H-adm, a FORCED statement does not depend on any member of $\mathcal{A}$ beyond the stated tolerance, so no declared input is smuggled into it. Every input dependence is either below tolerance or reported. The residual dependence is supported where Theorems 3.2 and 4.2 place it: on the observational kernel, before mixing, and on the phase weights of non-mixing dynamics.*

Proof. By definition. The localization follows from the two theorems cited. □

The certificate classified every case correctly (UG08):

statement	certificate	reason
identifiable posterior mean ($n = 10^5$)	FORCED	quotient, Bernstein–von Mises
kernel variance	CONDITIONAL	fibre: the prior is never forgotten
chaotic forecast spread, $n = 40$	FORCED	after mixing
chaotic forecast spread, $n = 3$	CONDITIONAL	before mixing
periodic forecast mean, wide input	CONDITIONAL	phase weights preserved

A certificate computed from one input law declares everything FORCED; the canary catches it. Selecting $\mathcal{A}$ after seeing the output would itself be smuggling, which is why H-adm requires the family to be declared in advance.

Corollary 7.4 (When Monte Carlo is correct; UG10). *Monte Carlo over an input law is objective exactly on FORCED outputs. It also needs its own validity certificate: the effective sample size must resolve the claimed differences. In the battery:*

- *two priors agree on a quotient output to 0.01–0.04 posterior standard deviations;*
- *they disagree on a kernel output by 43–61% in relative variance;*
- *the effective sample size is above the derived threshold of 200.*

8 How to compute uncertainty geometrically

1. *Locate.* Compute the kernel, the quotient and the strata (L1–L16).
2. *Regular strata.* Form the efficient Fisher information on the quotient (Schur complement over declared nuisance, L27). Its inverse gives the LAN ellipsoid. Check regularity with the rank certificate (L38).
3. *Walls.* Replace ellipsoids by preimage correspondence sets and tangent cones (L15–L19).
4. *Kernel directions.* Report the declared prior explicitly, as CONDITIONAL.
5. *Dynamics.* Before the spreading time, uncertainty is the propagated input, which is CONDITIONAL. After it, uncertainty is the invariant law (FORCED). Report the frontier (chaos paper).
6. *Quantum.* Use trace-rule probabilities and the task-selected metric.
7. *Monte Carlo.* Use it only on FORCED outputs, with an effective-sample-size certificate.

Why the methods sometimes match. Linearized propagation, Monte Carlo and the RG computation agree exactly where three conditions hold together:

- the stratum is regular (LAN; L22–L30, [9]);
- the requested quantity lies on the quotient;
- the dynamics has mixed.

They disagree at walls (branch selection), along the kernel (the prior is returned), and before mixing (the input is returned). The geometric computation says which situation applies.

9 Relation to the early framework

The early time-uncertainty framework and the CHI paper contained four correct intuitions:

- uncertainty must be propagated, not erased;
- a hazard index based on the conditioning of the Fisher matrix;
- stacking of local windows;
- the danger of “seepage” of regime information into flexible parameters.

In RG they become the rank certificate (L38), typed window certificates (L68), the non-circularity certificate (Section 7), and declared nuisance with efficient metrics (L27). The scalar compression and the single-anchor transport of CHI are not retained. A product of powers transports with one anchor only when the true law is that product (Buckingham, L70), and a constant Fisher rank cannot certify it.

10 Place in the ladder and non-claims

- *Placement.* The primitive and splitting theorem refine L1–L3 and L35–L37 (nulls). The locks refine L22–L30 (statistics) and L31–L34 (quantum). The certificates instantiate L87–L90 (closure).
- *Non-claims.* The locks are the cited theorems under their hypotheses. Non-regular, non-dominated or infinite-dimensional experiments are outside the LAN lock. The admissible family is a declaration and is not derived.

A Reproducibility

`rg_uncertainty_attack.py` passes 10/10 on seeds 20261009, 1, 42, 777 and 2026. The canaries are:

- a single input family in the certificate;
- a full-rank law, which leaves no kernel;
- a chaotic system substituted for the periodic control.

All are caught. Corrections before registration:

1. UG02 first measured TV with fixed bins, which are blind to narrow ensembles, and claimed a periodic system remembers every input; contraction collapses a narrow input, so the claim was restricted to phase weights.
2. UG03 first expected the kernel variance to be constant from $n = 10$; its limit is the conditional prior variance.
3. UG04 first tested near the fold, where large standard errors mask the failure; the decisive case is branch selection.
4. UG08 first used an absolute tolerance; it is now relative to the output’s scale.
5. UG10 first used $n = 5000$, where importance weights collapsed; it now carries an effective-sample-size gate derived from the required resolution.

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